SIMATS ENGINEERING

Lab Programs

COURSE CODE: ECA1009 **COURSE NAME: Microprocessors & Microcontrollers**

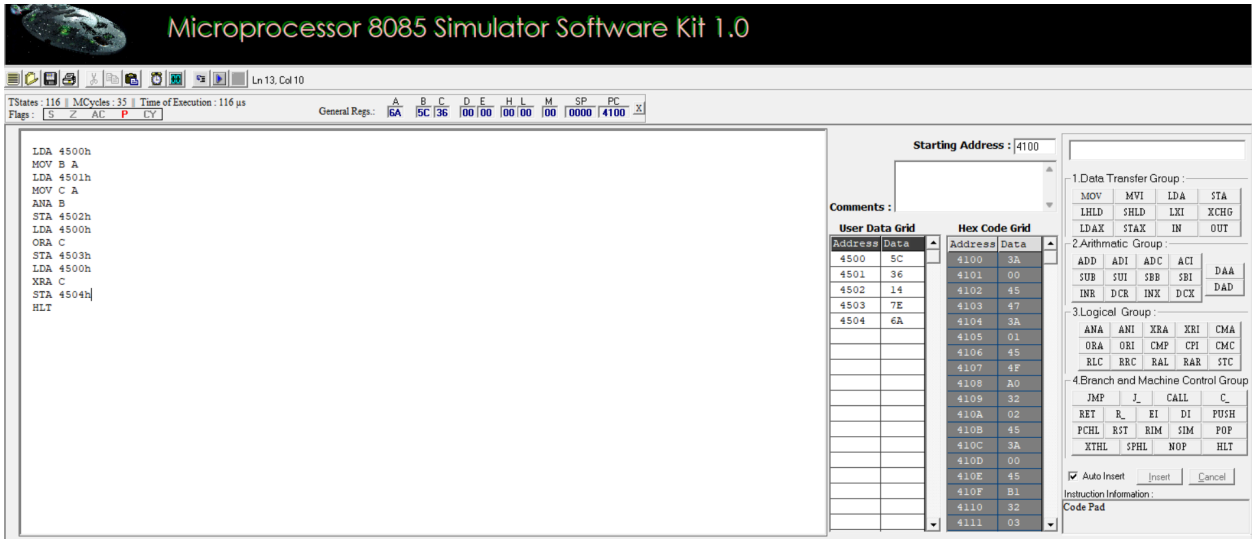
Program A – Verification of Bit Masking Operations Using 8-Bit Processor Assembly Language Programming

Aim: To implement and verify bit masking operations using 8-bit Intel 8085 assembly language programming.

Algorithm:

1. Load the first 8-bit data from memory location 4500H into the accumulator.
2. Transfer the data to register B.
3. Load the second 8-bit data from memory location 4501H into the accumulator and transfer it to register C.
4. Perform the AND operation using `ANA B` and store the result at 4502H.
5. Perform the OR operation using `ORA C` and store the result at 4503H.
6. Perform the XOR operation using `XRA C` and store the result at 4504H, then halt the program.

Execute:



Result: The bit masking operations AND, OR, and XOR were executed successfully using the 8085 simulator, and the results were stored in the specified memory locations.